

UNIT – 2  
ASSESSMENT – 3  
**OUTPUT**

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1.

A\* SEARCH ALGORITHM

1. Use Faculty Graph
2. Enter Custom Graph

Enter Choice : 5

Enter Number of Nodes : 4

Node Name : A

Node Name : B

Node Name : C

Node Name : D

Enter Number of Edges : 6

Edge 1

From : A

To : D

Cost : 5

Edge 2

From : A

To : C

Cost : 3

Edge 3

From : D

To : A

Cost : 6

Edge 4

From : D

To : B

Cost : 5

2.

Beta = 6  
Current Minimum = 6

Leaf Node : 4  
Alpha = 2  
Beta = 4  
Current Minimum = 4

RIGHT MIN returns : 4

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Iteration 3  
Evaluating ROOT MAX  
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Left Value : 2  
Right Value : 4

MAX chooses : 4

===== FINAL RESULT =====  
Best Move : RIGHT Subtree  
Final Minimax Value : 4  
Pruned Nodes : None

===== MINIMAX WITH ALPHA-BETA PRUNING =====

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Iteration 1  
Evaluating LEFT MIN Node  
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Leaf Node : 5  
Alpha = -inf  
Beta = 5  
Current Minimum = 5

|  
Leaf Node : 3  
Alpha = -inf  
Beta = 3  
Current Minimum = 3

Leaf Node : 2  
Alpha = -inf  
Beta = 2  
Current Minimum = 2

LEFT MIN returns : 2

MAX updates Alpha = 2

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Iteration 2  
Evaluating RIGHT MIN Node  
-----

Leaf Node : 9  
Alpha = 2  
Beta = 9  
Current Minimum = 9

Leaf Node : 6  
Alpha = 2

Beta = 6  
Current Minimum = 6

Leaf Node : 4  
Alpha = 2  
Beta = 4  
Current Minimum = 4

RIGHT MIN returns : 4

-----  
Iteration 3  
Evaluating ROOT MAX  
-----

Left Value : 2  
Right Value : 4

MAX chooses : 4

===== FINAL RESULT =====  
Best Move : RIGHT Subtree  
Final Minimax Value : 4  
Pruned Nodes : None